Deployment summary

NetApp Solutions

Dorian Henderson
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For the initial validation of this solution, NetApp partnered with World Wide Technology (WWT) to establish an environment at WWT's Advanced Technology Center (ATC). Anthos was deployed on a bare metal infrastructure using the bmctl tool provided by Google Cloud. The following section details the deployment used for validation purposes.

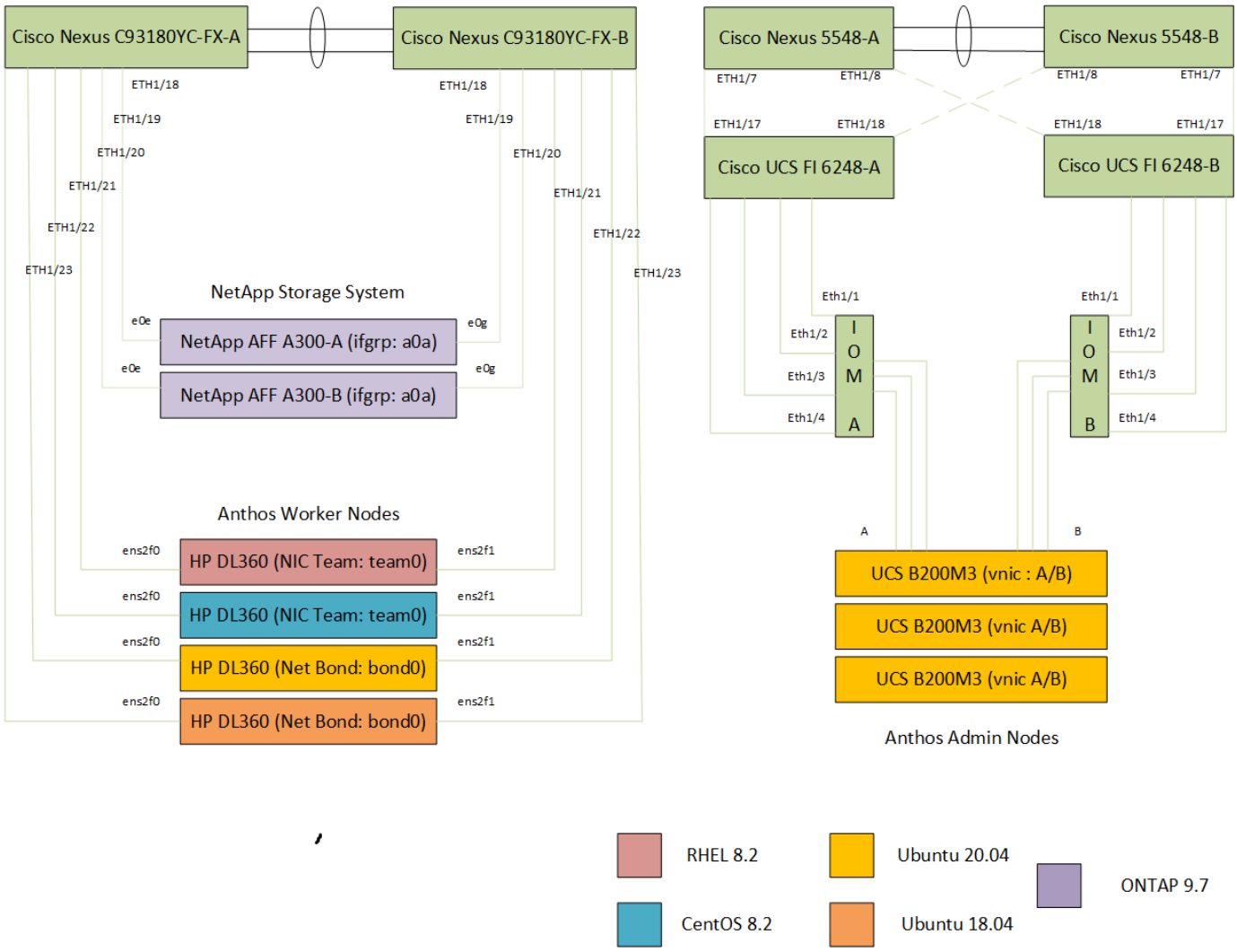
The Anthos on bare metal with NetApp solution was built as a highly available hybrid cluster with three Anthos control-plane nodes and four Anthos worker nodes.

The control-plane nodes used were Cisco UCS B200M3 blade servers hosted in a chassis and configured with a single virtual network interface card (vNIC) on each, which allowed for A/B failover at the Cisco UCS platform level for fault tolerance. The Cisco UCS chassis connected upstream to a pair of Cisco UCS 6248 fabric interconnects providing disparate paths for the separation of traffic along fabric A and fabric B. Those fabric interconnects connected upstream to a pair of Cisco Nexus 5548 data center switches that tied back to the core network at WWT.

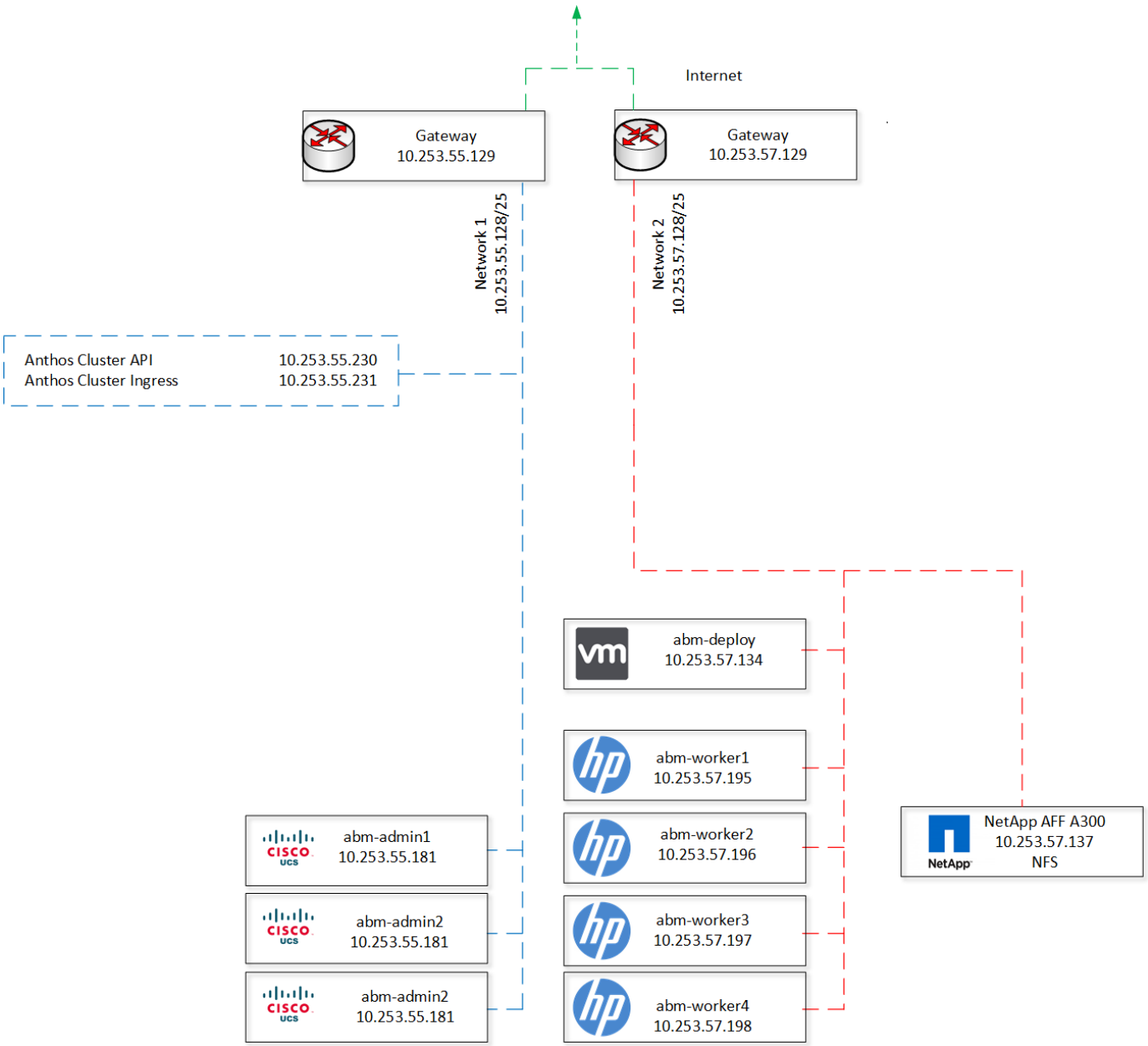
The worker nodes were HP Proliant DL360 nodes, each running one of the supported Linux distributions for Anthos on bare metal: Red Hat Enterprise Linux 8.2, CentOS 8.2, Ubuntu 20.04 LTS, or Ubuntu 18.04 LTS. The Red Hat Enterprise Linux 8 and CentOS 8 nodes were configured with NIC teams running in LACP mode and cabled to two Nexus 9k C93180YC-FX switches for fault tolerance. The Ubuntu servers were configured for network bonding in LACP mode and cabled to the same pair of Nexus 9k switches for fault tolerance.

The NetApp AFF A300 storage system running ONTAP 9.7 software was installed and connected physically to the same pair of Nexus 9k switches as the Anthos worker nodes. These network uplinks were aggregated into an interface group (a0a), and the appropriate data network VLAN was tagged to allow the worker nodes to interact with the storage system. A storage virtual machine (SVM) was created with data LIFs supporting the NFS protocol and dedicated to storage operations for Trident to provide persistent storage to the containers deployed in the Anthos on bare metal cluster. These persistent volumes were provided by NetApp Trident 20.10, the latest release of the fully supported NetApp open-source storage orchestrator for Kubernetes.

The following figure depicts a physical cabling diagram of the solution to the top of rack data center switches.



The next figure presents a logical view of the solution as deployed and validated on the hardware in the lab at the NetApp partner WWT.



Next: Solution validation.

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